Data Preparation

Date	20 September 2026
Team ID	Roll 58
Project Name	EV Market Analysis Dashboard using Tableau Public
Maximum Marks	5 Marks

Dataset: ev_market_train.csv
Dataset Size: 200,000 records and 37 columns

Introduction
Data preparation is the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting.
For this project, the Electric Vehicle (EV) dataset contains information about manufacturers, vehicle categories, battery technologies, driving range, price, charging performance, customer ratings, manufacturing countries, and annual sales. Data preparation will identify missing values, handle duplicate records, standardize inconsistent values, and create useful derived fields for the EV Market Analysis Dashboard.
The prepared dataset will be used to build reliable visualizations in Tableau Public.
Snowflake table assumption: The source data has been loaded into RAW.EV_MARKET_TRAIN. Replace this table name if your actual Snowflake table has a different name.

Activity 1: Identifying and Handling NULL Values

Objective
Identify NULL or missing values in the EV dataset, determine which fields are affected, and decide how to handle missing information without introducing misleading results.

SQL Code 1: Identify NULL values in all columns

```
SELECT 'VEHICLE_ID' AS COLUMN_NAME,
COUNT_IF(VEHICLE_ID IS NULL) AS NULL_COUNT
FROM RAW.EV_MARKET_TRAIN

UNION ALL
SELECT 'RECOMMENDED_USE_CASE',
COUNT_IF(RECOMMENDED_USE_CASE IS NULL)
FROM RAW.EV_MARKET_TRAIN
....
ORDER BY NULL_COUNT DESC;
```

Output

The query returns two columns:

COLUMN_NAME	NULL_COUNT
Column containing missing values	Actual count from Snowflake
Another affected column	Actual count from Snowflake
Remaining columns	Actual counts from Snowflake

A count of 0 means that no SQL NULL values were found in that column. Empty text strings should also be checked because they are not necessarily stored as NULL.

SQL Code 2: Check empty text fields

```
SELECT
COUNT_IF(TRIM(MODEL_NAME) = '') AS BLANK_MODEL_NAMES,
COUNT_IF(TRIM(MANUFACTURER) = '') AS BLANK_MANUFACTURERS,
COUNT_IF(TRIM(CATEGORY) = '') AS BLANK_CATEGORIES,
```

```
COUNT_IF(TRIM(BATTERY_TYPE) = '') AS BLANK_BATTERY_TYPES,
COUNT_IF(TRIM(MANUFACTURING_COUNTRY) = '')
AS BLANK_COUNTRIES
FROM RAW.EV_MARKET_TRAIN;
```

Handling strategy

- Retain missing values in optional technical fields when filling them would introduce unsupported information.
- Investigate missing ANNUAL_SALES_UNITS values before calculating total sales.
- Investigate missing PRICE_USD, RANGE_KM, and CUSTOMER_RATING values before using these fields in comparisons.
- Avoid replacing missing values with zero unless zero has a valid business meaning.
- Convert empty strings in text fields into NULL values during staging.

Activity 2: Duplicate Record Handling

Objective

Identify exact duplicate rows and repeated vehicle identifiers. Determine which records can be removed without losing legitimate information.

SQL Code 1: Count exact duplicate rows

```
SELECT
(SELECT COUNT(*)
FROM RAW.EV_MARKET_TRAIN) AS TOTAL_ROWS,

(SELECT COUNT(*)
FROM (
SELECT DISTINCT *
FROM RAW.EV_MARKET_TRAIN
)) AS DISTINCT_ROWS,

(SELECT COUNT(*)
FROM RAW.EV_MARKET_TRAIN)
-
(SELECT COUNT(*)
FROM (
SELECT DISTINCT *
FROM RAW.EV_MARKET_TRAIN
)) AS EXACT_DUPLICATE_ROWS;
```

Output

TOTAL_ROWS	DISTINCT_ROWS	EXACT_DUPLICATE_ROWS
Actual count	Actual count	Actual count

EXACT_DUPLICATE_ROWS represents the number of rows beyond the distinct set of complete records.

SQL Code 2: Find repeated vehicle identifiers

```
SELECT
VEHICLE_ID,
COUNT(*) AS RECORD_COUNT
FROM RAW.EV_MARKET_TRAIN
GROUP BY VEHICLE_ID
HAVING COUNT(*) > 1
ORDER BY RECORD_COUNT DESC;
```

Output

The query returns VEHICLE_ID and RECORD_COUNT for identifiers appearing more than once. If no records are returned, no repeated non-NULL vehicle identifiers were found.

SQL Code 3: Remove exact duplicate rows

```
CREATE OR REPLACE TABLE STAGING.EV_MARKET_DEDUP AS
SELECT DISTINCT *
FROM RAW.EV_MARKET_TRAIN;
```

Validation Code

```
SELECT
(SELECT COUNT(*)
FROM RAW.EV_MARKET_TRAIN) AS RAW_ROW_COUNT,

(SELECT COUNT(*)
FROM STAGING.EV_MARKET_DEDUP) AS DEDUP_ROW_COUNT;
```

Result and explanation

The STAGING.EV_MARKET_DEDUP table retains one copy of each distinct complete record. Its row count should be less than or equal to the original count.

Repeated VEHICLE_ID values must be reviewed separately. They should not automatically be deleted if they identify legitimate records with different specifications or other meaningful differences.

Activity 3: Data Cleaning and Standardization

Objective

Correct formatting inconsistencies, remove unwanted whitespace, convert empty text into NULL values, and ensure that numerical fields are suitable for calculation.

SQL Code 1: Create a cleaned staging table

```
CREATE OR REPLACE TABLE STAGING.EV_MARKET_TRAIN_CLEAN AS
SELECT
NULLIF(TRIM(TO_VARCHAR(VEHICLE_ID)), '') AS VEHICLE_ID,
NULLIF(TRIM(TO_VARCHAR(MODEL_NAME)), '') AS MODEL_NAME,
NULLIF(TRIM(TO_VARCHAR(MANUFACTURER)), '') AS MANUFACTURER,
…
```

SQL Code 2: Validate cleaned fields

```
SELECT
COUNT(*) AS TOTAL_RECORDS,

COUNT_IF(VEHICLE_ID IS NULL) AS MISSING_VEHICLE_ID,

COUNT_IF(MANUFACTURER IS NULL) AS MISSING_MANUFACTURER,

COUNT_IF(ANNUAL_SALES_UNITS IS NULL)
AS MISSING_SALES,

COUNT_IF(PRICE_USD IS NULL) AS MISSING_PRICE,

COUNT_IF(RANGE_KM IS NULL) AS MISSING_RANGE,

COUNT_IF(CUSTOMER_RATING IS NULL)
AS MISSING_CUSTOMER_RATING

FROM STAGING.EV_MARKET_TRAIN_CLEAN;
```

Output

TOTAL_RECORDS	MISSING_SALES	MISSING_PRICE	MISSING_RANGE	MISSING_CUSTOMER_RATING
Actual count	Actual count	Actual count	Actual count	Actual count

The results help determine whether the cleaning process introduced additional missing values and whether the critical fields are sufficiently complete for analysis.

Activity 4: Data Transformation and Validation

Objective

Transform the cleaned EV dataset into a curated table suitable for dashboard analysis. Create useful derived fields and validate the business measures before using them in Tableau Public.

SQL Code 1: Create derived fields

```
CREATE OR REPLACE TABLE CURATED.EV_MARKET_TRAIN AS
SELECT
S.*,ROUND(
PRICE_USD / NULLIF(RANGE_KM, 0),
2
) AS PRICE_USD_PER_KM_RANGE,
CASE
WHEN RANGE_KM IS NULL OR RANGE_KM < 0
AND (CUSTOMER_RATING < 0 OR CUSTOMER_RATING > 5)
THEN 'Check Rating'
ELSE 'Review Passed'
END AS DATA_QUALITY_STATUS

FROM STAGING.EV_MARKET_TRAIN_CLEAN S;
```

The range and price bands are project-defined categories for easier comparison. They are not official industry classifications. The data-quality status flags suspicious values for review; it does not automatically delete them.

SQL Code 2: Validate the curated dataset

```
SELECT
COUNT(*) AS CURATED_RECORDS,

COUNT_IF(PRICE_USD <= 0)
AS NONPOSITIVE_PRICES,
COUNT_IF(RANGE_KM < 0)
AS NEGATIVE_RANGES,
COUNT_IF(ANNUAL_SALES_UNITS < 0)
AS NEGATIVE_SALES,
COUNT_IF(
CUSTOMER_RATING < 0 OR CUSTOMER_RATING > 5
) AS INVALID_CUSTOMER_RATINGS,
....
COUNT_IF(DATA_QUALITY_STATUS <> 'Review Passed')
AS RECORDS_REQUIRING_REVIEW

FROM CURATED.EV_MARKET_TRAIN;
```

Output

CURATED_RECORDS	NONPOSITIVE_PRICES	NEGATIVE_RANGES	NEGATIVE_SALES	INVALID_CUSTOMER_RATINGS	RECORDS_REQUIRING_REVIEW
Actual count	Actual count	Actual count	Actual count	Actual count	Actual count

These output values must be copied from the actual query results. NULL values are excluded from ordinary comparisons such as PRICE_USD <= 0, so missing-value checks from Activity 1 should also be retained.

SQL Code 3: Final validation for Tableau Public

```
SELECT
MANUFACTURER,
CATEGORY,

SUM(ANNUAL_SALES_UNITS) AS TOTAL_RECORDED_SALES,

ROUND(AVG(RANGE_KM), 2) AS AVERAGE_RANGE_KM,

ROUND(AVG(PRICE_USD), 2) AS AVERAGE_PRICE_USD,

ROUND(AVG(CUSTOMER_RATING), 2) AS AVERAGE_CUSTOMER_RATING

FROM CURATED.EV_MARKET_TRAIN

GROUP BY MANUFACTURER, CATEGORY

ORDER BY TOTAL_RECORDED_SALES DESC;
```

Expected result structure

The result provides one row per manufacturer and category combination, with aggregated sales, average range, average price, and average customer rating. It can be used to verify the values shown in the dashboard.

Final Outcome

The data preparation process creates a structured workflow from the RAW dataset to the STAGING and CURATED layers. Missing values are profiled, exact duplicates are handled, formatting and data types are standardized, derived fields are created, and quality checks are performed.

The curated EV dataset is then ready for use in Tableau Public visualizations, including sales comparisons, battery analysis, range-versus-price charts, customer rating analysis, and vehicle-category comparisons.

Note: All actual counts and validation results must be taken from Snowflake after running the SQL statements. The dataset should not be described as fully clean until those results have been reviewed.